

[GETNET] - Kit de Diagnóstico Oracle - Sessões, Locks e Flashback

===== KIT DE DIAGNÓSTICO ORACLE - SESSÕES, LOCKS E FLASHBACK =====

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 Ambiente: Oracle Database (RAC ou Single)
 Finalidade: Diagnóstico de sessões, bloqueios, waits e atividades de Flashback Archive.
 Compatibilidade: Oracle 11g, 12c, 19c, 21c

===== SEÇÕES DESTE ARQUIVO =====

- 1) IDENTIFICAÇÃO DE ATIVIDADES RECENTES (ASH)
- 2) HISTÓRICO DE SESSÃO (AWR)
- 3) TEXTOS DE SQL EXECUTADOS
- 4) PLANO DE EXECUÇÃO DE SQL
- 5) RESUMO DE EVENTOS E OBJETOS TOCADOS
- 6) IDENTIFICAÇÃO DE SESSÕES BLOQUEADORAS E BLOQUEADAS
- 7) MONITORAMENTO DE ROLLBACK / TRANSAÇÕES LONGAS
- 8) VERIFICAÇÃO DE FLASHBACK DATA ARCHIVE
- 9) DICAS E BOAS PRÁTICAS

----- 1) IDENTIFICAÇÃO DE ATIVIDADES RECENTES (ASH) -----

```
-- Mostra o que a sessão estava fazendo nos últimos minutos.
-- Usa V$ACTIVE_SESSION_HISTORY (janela em memória ~1h)
-- Substitua &SID pelo número da sessão e ajuste &MIN (tempo em minutos).
set lines 300 pages 200 trims on tab off
col sample_time      for a20
col user_prog        for a35
col module_action    for a35
col event            for a40
col obj              for a40
col sql_id           for a13
col plan_operation   for a35
col wait_class       for a15
col machine          for a25
col client_id        for a20
define SID=1246
define MIN=60 -- minutos de janela
select
  to_char(sample_time,'YYYY-MM-DD HH24:MI:SS') as sample_time,
  session_id as sid,
  session_serial# as serial#,
  sql_id,
  nvl(event, session_state) as event,
  wait_class,
  module||' / '||action as module_action,
  nvl(program,'-') as user_prog,
  nvl(machine,'-') as machine,
  client_id,
  decode(current_obj#, -1, '-', obj.owner||'.'||obj.object_name||' ('||obj.object_type||')') as obj,
  plan_operation
from v$active_session_history ash
left join dba_objects obj on obj.object_id = ash.current_obj#
where session_id = &SID
  and sample_time >= systimestamp - numtodsinterval(&MIN,'minute')
order by sample_time;
```

----- 2) HISTÓRICO DE SESSÃO (AWR) -----

```
-- Mostra atividades passadas de uma sessão específica (se já saiu da memória)
-- Necessita Diagnostics Pack habilitado.
set lines 300 pages 200 trims on tab off
col sample_time for a20
col event for a40
col obj for a40
col module_action for a35
col user_prog for a35
col wait_class for a15
col sql_id for a13
col inst for 999
define SID=1246
define DT_INI='2025-10-05 00:00:00'
define DT_FIM='2025-10-05 23:59:59'
with rng as (
  select to_date('&DT_INI','YYYY-MM-DD HH24:MI:SS') t1,
         to_date('&DT_FIM','YYYY-MM-DD HH24:MI:SS') t2
  from dual
)
```

```

select
  to_char(sample_time,'YYYY-MM-DD HH24:MI:SS') as sample_time,
  inst_id as inst,
  session_id as sid,
  session_serial# as serial#,
  sql_id,
  nvl(event, session_state) as event,
  wait_class,
  module||' / '||action as module_action,
  nvl(program,'') as user_prog,
  decode(current_obj#, -1, '-', obj.owner||'.'||obj.object_name||' ('||obj.object_type||')') as obj,
  blocking_session as blk_sid
from dba_hist_active_sess_history ash
left join dba_objects obj on obj.object_id = ash.current_obj#
cross join rng
where session_id = &SID
      and sample_time between rng.t1 and rng.t2
order by sample_time;
=====

```

3) TEXTOS DE SQL EXECUTADOS

```

-- Mostra o SQL completo, estatísticas de execução e o texto do comando.
-- Use o SQL_ID obtido no ASH/AWR.
set long 100000 longchunksize 100000 lines 300 pages 100
col sql_fulltext for a200
define SQLID='&SQL_ID_AQUI'
select sql_id, parsing_schema_name, module, first_load_time, executions,
       elapsed_time/1e6 elapsed_s, cpu_time/1e6 cpu_s, buffer_gets, disk_reads
from v$sql
where sql_id = '&SQLID';
select sql_fulltext
from v$sql
where sql_id = '&SQLID';
select sql_id, dbid, to_char(last_active_time,'YYYY-MM-DD HH24:MI:SS') last_active_time
from dba_hist_sqlstat
where sql_id = '&SQLID'
order by last_active_time desc;
select sql_text
from dba_hist_sqltext
where sql_id = '&SQLID';
=====

```

4) PLANO DE EXECUÇÃO DE SQL

```

-- Mostra as operações do plano (TABLE ACCESS, JOIN, SORT, etc.)
set lines 300 pages 200
col operation for a35
col options for a25
col object_name for a35
select id, parent_id, operation, options, object_owner, object_name, cardinality, cost
from v$sql_plan
where sql_id = '&SQLID'
order by id;
-- Histórico (AWR)
select id, parent_id, operation, options, object_owner, object_name, cardinality, cost
from dba_hist_sql_plan
where sql_id = '&SQLID'
order by id;
=====

```

5) RESUMO DE EVENTOS E OBJETOS TOCADOS

```

-- Agrupa os principais waits e objetos acessados.
set lines 300 pages 200
col event for a40
col wait_class for a20
define SID=1246
define DT_INI='2025-10-05 00:00:00'
define DT_FIM='2025-10-05 23:59:59'
select
  nvl(event, session_state) as event,
  wait_class,
  count(*) samples,
  round(count(*) * 1.0 / (select count(*) from dba_hist_active_sess_history
                        where session_id=&SID
                        and sample_time between to_date('&DT_INI','YYYY-MM-DD HH24:MI:SS')
                        and to_date('&DT_FIM','YYYY-MM-DD HH24:MI:SS'))*100,2) pct
from dba_hist_active_sess_history
where session_id = &SID
      and sample_time between to_date('&DT_INI','YYYY-MM-DD HH24:MI:SS')
      and to_date('&DT_FIM','YYYY-MM-DD HH24:MI:SS')
group by nvl(event, session_state), wait_class
order by samples desc;
col obj for a60
select
  decode(current_obj#, -1, '-', obj.owner||'.'||obj.object_name||' ('||obj.object_type||')') as obj,
  count(*) samples
from dba_hist_active_sess_history ash
left join dba_objects obj on obj.object_id = ash.current_obj#

```

```

where session_id = &SID
and sample_time between to_date('&DT_INI','YYYY-MM-DD HH24:MI:SS')
and to_date('&DT_FIM','YYYY-MM-DD HH24:MI:SS')
group by decode(current_obj#, -1, '-', obj.owner||'.'||obj.object_name||' ('||obj.object_type||')')
order by samples desc nulls last;
=====

6) IDENTIFICAÇÃO DE SESSÕES BLOQUEADORAS E BLOQUEADAS
-----
-- Mostra quem está bloqueando quem, objetos e status.
col username for a20
select s.sid, s.serial#, s.username, s.status, s.blocking_session "BLOCKED_BY"
from v$session s
where s.blocking_session is not null
order by s.sid;
-- Sessões e objetos bloqueados
select s.sid, s.serial#, s.username, o.owner||'.'||o.object_name objeto
from v$session s
join v$locked_object lo on s.sid=lo.session_id
join dba_objects o on o.object_id=lo.object_id;
=====

7) MONITORAMENTO DE ROLLBACK / TRANSAÇÕES LONGAS
-----
-- Verifica rollback pendente após matar sessões.
col username for a20
col status for a10
select s.sid, s.serial#, s.username, s.status, t.used_urec undo_records, t.used_ublk undo_blocks
from v$session s join v$transaction t on s.saddr=t.ses_addr
order by t.used_ublk desc;
col opname for a40
select sid, sofar, totalwork, units, elapsed_seconds, time_remaining, opname
from v$session_longops
where sofar < totalwork;
=====

8) VERIFICAÇÃO DE FLASHBACK DATA ARCHIVE
-----
-- Mostra tabelas com Flashback Archive habilitado e uso de espaço.
select p.pid, p.spid, s.sid, s.serial#, s.program
from v$process p join v$session s on p.addr=s.paddr
where s.program like '%(FBDA)%';
select owner, table_name, flashback_archive_name, retention_in_days
from dba_flashback_archive_tables;
select name, retention, space_limit, space_used
from dba_flashback_archive;
=====

9) DICAS E BOAS PRÁTICAS
-----
✓ Sempre identifique o SQL_ID antes de matar uma sessão
✓ Mate apenas sessões de usuário, nunca processos de background (SMON, PMON, LGWR etc.)
✓ Após matar, monitore rollback (v$transaction)
✓ Agende purges de Flashback Archive fora do horário de pico
✓ Se o lock for recorrente, considere:
  - Reduzir RETENTION do Flashback Archive
  - Desativar temporariamente o Flashback em tabelas de batch
  - Isolar o Flashback Archive em tablespace próprio
✓ Salve os relatórios para comparação histórica (CPU, waits, tempo)
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FIM DO ARQUIVO
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```